Quantum Reference Frames

Relational states, frame changes and operational frames

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Chapter 0 Preliminaries

Standing assumptions: *Unless stated otherwise, G is a* finite *group, with averages $\int dg := \frac{1}{|G|}\sum_{g\in G}$, and all Hilbert spaces are finite-dimensional. Remarks indicate the extension to compact and locally compact groups. We write $\mathcal{U}_g(X) = U(g)XU(g)^\dagger$.*

0.1 Representations and the twirl

Definition 0.1.1 (Charge sectors): A unitary representation U on $\mathcal{H}$ decomposes as

$$\mathcal{H} = \bigoplus_q \mathcal{M}_q \otimes \mathcal{N}_q, \qquad U(g) = \bigoplus_q U_q(g) \otimes \mathbb{I}_{\mathcal{N}_q},$$

with U_q irreducible on the gauge spaces $\mathcal{M}_q$ and multiplicity spaces $\mathcal{N}_q$.

Definition 0.1.2 (Twirl, invariance, covariance):

- $\mathcal{G}(X) = \int dg\, \mathcal{U}_g(X)$.
- X is *invariant* if $\mathcal{U}_g(X) = X$ for all g.
- A channel $\mathcal{E}$ is *covariant* if $\mathcal{U}_g^B \circ \mathcal{E} = \mathcal{E} \circ \mathcal{U}_g^A$ for all g.

Theorem 0.1.1: Structure of the twirl

$$\mathcal{G} = \sum_q (\mathcal{D}_{\mathcal{M}_q} \otimes \mathcal{I}_{\mathcal{N}_q}) \circ \mathcal{P}_q,$$

where $\mathcal{P}_q(X) = \Pi_q X \Pi_q$ and $\mathcal{D}$ is the complete depolariser. The invariant operators are exactly $\bigoplus_q \mathbb{I}_{\mathcal{M}_q} \otimes X_q$.

Proof. Apply Schur's lemma to each block $\int dg\, U_q(g) B U_{q'}(g)^\dagger$, which is an intertwiner by invariance of dg:

- it vanishes for $q \neq q'$;
- it equals $(\operatorname{Tr} B/d_q)\mathbb{I}$ for $q = q'$.

Invariant X satisfy $X = \mathcal{G}(X)$. □

0.2 External frames and the invariance principle

Theorem 0.2.1: Lack of an external frame = superselection rule

Suppose each party can implement any state, measurement or channel *relative to its own frame*, and the two frames are related by a completely unknown g (Haar prior). Relative to one party, the other can then realise exactly

- the invariant states,
- the POVMs with invariant effects,
- the covariant channels.

Proof. Averaging $\mathcal{U}_g(\rho)$, $\mathcal{U}_g(E)$ and $\mathcal{U}_g \circ \mathcal{E} \circ \mathcal{U}_{g^{-1}}$ over g gives invariant or covariant objects. Conversely, these objects are fixed by the averaging, so each of them is realised by preparing, measuring or applying it relative to one's own frame. □

Definition 0.2.1 (Partially shared external frame): If g is distributed according to μ, the effective description is the weighted twirl $\mathcal{G}_\mu = \int d\mu(g)\, \mathcal{U}_g$.

- $\mu = \delta_e$: perfectly shared frame.
- $\mu =$ Haar: no shared frame.

0.3 Two notions of symmetric state

Definition 0.3.1 (Invariant vs. physical states): For a representation U on $\mathcal{H}$, with $\Pi_0 = \int dg\, U(g)$ the projector onto the invariant vectors (the trivial-charge sector), define

$$\begin{aligned} \mathcal{S}_{\text{inv}} &= \{\rho : \mathcal{G}(\rho) = \rho\} && \text{(invariant density operators)}, \\ \mathcal{S}_{\text{phys}} &= \{\rho : \Pi_0 \rho\, \Pi_0 = \rho\} && \text{(density operators on invariant vectors)}. \end{aligned}$$

Lemma 0.3.2: $\mathcal{S}_{\text{phys}} \subsetneq \mathcal{S}_{\text{inv}}$ in general. By Theorem 0.1.1, $\mathcal{S}_{\text{inv}} = \{\bigoplus_q \mathbb{I}/d_q \otimes \rho_q\}$ allows weight in every charge sector, whereas $\mathcal{S}_{\text{phys}}$ is the trivial-charge sector alone.

Proof. If $\rho = \Pi_0 \rho \Pi_0$, then $U(g)\rho U(g)^\dagger = U(g)\Pi_0 \rho \Pi_0 U(g)^\dagger = \Pi_0 \rho \Pi_0 = \rho$, using $U(g)\Pi_0 = \Pi_0$.

Strictness: take $G = \mathbb{Z}_2$ acting by $X \otimes X$ on two qubits and $\rho = \frac{1}{2}(|00\rangle\langle 00| + |11\rangle\langle 11|)$. Then $\rho \in \mathcal{S}_{\text{inv}}$, but $\text{Tr}[\Pi_0 \rho] = \frac{1}{2}(1 + \text{Tr}[(X \otimes X)\rho]) = \frac{1}{2}$. □

Remark 0.3.3: The two sets underlie different approaches:

- $\mathcal{S}_{\text{inv}}$: the quantum-information approach without an external frame (*incoherent* group averaging, Chapter 0.2, Chapter 4).
- $\mathcal{S}_{\text{phys}}$: the perspective-neutral constraint approach (*coherent* group averaging, Chapters 1–2).

Both are self-consistent, but they are not equivalent as state spaces. Proposition 1.3.2 shows only this: the coherent and incoherent embeddings of the *same* system state yield identical statistics for the algebra of relational observables, although the resulting global states are generally distinct. It does *not* assert a correspondence between all of $\mathcal{S}_{\text{inv}}$ and $\mathcal{S}_{\text{phys}}$.

Invariance principle. If there is no external frame at all, only *invariant* quantities are physical. Non-invariant ("absolute") quantities must then be recovered as *relations* between the system and another physical system, the quantum reference frame. Chapters 1–3 make this precise.

Chapter 1 Quantum Reference Frames: the Perspective-Neutral Picture

1.1 Ideal quantum reference frames

Definition 1.1.1 (Ideal QRF): An *ideal QRF* for G is the system R with

$$\mathcal{H}_R = \mathbb{C}^{|G|} = \text{span}\{|g\rangle\}_{g \in G}, \qquad U_R(h)|g\rangle = |hg\rangle$$

(the left-regular representation). The vectors $|g\rangle$ are the *orientation states*; they are orthonormal, hence perfectly distinguishable.

Definition 1.1.2 (Kinematical and physical Hilbert spaces): For a frame R and a system S with representation U_S:

- the kinematical space is $\mathcal{H}_{\text{kin}} = \mathcal{H}_R \otimes \mathcal{H}_S$, carrying the global action $U(g) = U_R(g) \otimes U_S(g)$;
- the physical (perspective-neutral) space is $\mathcal{H}_{\text{phys}} = \{\Psi : U(g)\Psi = \Psi\ \forall g\} = \Pi_{\text{phys}} \mathcal{H}_{\text{kin}}$, with

$$\Pi_{\text{phys}} = \int dg\, U_R(g) \otimes U_S(g).$$

Lemma 1.1.3: Π_{phys} is the orthogonal projector onto the invariant vectors.

Proof. $U(h)\Pi_{\text{phys}} = \Pi_{\text{phys}}$ by invariance of the average, so $\Pi_{\text{phys}}^2 = \Pi_{\text{phys}}$. Also $\Pi_{\text{phys}}^\dagger = \Pi_{\text{phys}}$, since $g \mapsto g^{-1}$ preserves the average. Invariant Ψ satisfy $\Pi_{\text{phys}}\Psi = \Psi$. □

Remark 1.1.4: $\mathcal{H}_{\text{kin}}$ still refers to a fictitious external frame. $\mathcal{H}_{\text{phys}}$ contains only gauge-invariant information: it is the description "from no particular perspective".

1.2 Reduction to the perspective of a frame

Lemma 1.2.1 (Form of physical states): $\Psi \in \mathcal{H}_{\text{phys}}$ iff

$$\Psi = \frac{1}{\sqrt{|G|}} \sum_{g\in G} |g\rangle_R \otimes U_S(g)\psi$$

for a (unique) $\psi \in \mathcal{H}_S$. Then $\|\Psi\| = \|\psi\|$.

Proof. Write $\Psi = \sum_g |g\rangle \otimes \psi_g$. Then

$$U(h)\Psi = \sum_g |hg\rangle \otimes U_S(h)\psi_g,$$

so invariance means $\psi_{hg} = U_S(h)\psi_g$ for all g, h, i.e. $\psi_g = U_S(g)\psi_e$. Put $\psi = \sqrt{|G|}\,\psi_e$. The norm follows from $\|\Psi\|^2 = \sum_g \|U_S(g)\psi_e\|^2 = |G|\,\|\psi_e\|^2$. □

Theorem 1.2.1: Reduction map

The map

$$\mathcal{R}_e : \mathcal{H}_{\text{phys}} \to \mathcal{H}_S, \qquad \mathcal{R}_e\Psi := \sqrt{|G|}\,(\langle e|_R \otimes \mathbb{I}_S)\Psi,$$

is unitary, with inverse

$$\mathcal{R}_e^{-1}\psi = \frac{1}{\sqrt{|G|}} \sum_g |g\rangle \otimes U_S(g)\psi = \sqrt{|G|}\,\Pi_{\text{phys}}(|e\rangle \otimes \psi).$$

More generally, $\mathcal{R}_g := \sqrt{|G|}\,(\langle g| \otimes \mathbb{I})$ restricted to $\mathcal{H}_{\text{phys}}$ satisfies $\mathcal{R}_g = U_S(g)\,\mathcal{R}_e$.

Proof. By the lemma, $\mathcal{R}_e$ maps $\Psi \mapsto \psi$ bijectively and isometrically. For the second formula of the inverse, compute

$$\Pi_{\text{phys}}(|e\rangle \otimes \psi) = \frac{1}{|G|} \sum_g |g\rangle \otimes U_S(g)\psi.$$

Finally, $\sqrt{|G|}\,\langle g|\Psi = \sqrt{|G|}\,U_S(g)\psi_e = U_S(g)\psi$. □

Definition 1.2.2 (Relational state): $\mathcal{R}_e\Psi \in \mathcal{H}_S$ is the state of S *relative to* R (with R in orientation e). For mixed physical states ϱ on $\mathcal{H}_{\text{phys}}$, it is $\mathcal{R}_e\varrho\,\mathcal{R}_e^\dagger$.

Remark 1.2.3:

- $\dim \mathcal{H}_{\text{phys}} = \dim \mathcal{H}_S$: the frame's own degrees of freedom are "used up" in defining the perspective.
- The usual description of S relative to an external laboratory is recovered as the description relative to an ideal QRF in orientation e: $\mathcal{R}_e^{-1}\psi$ is the gauge-invariant completion of $|e\rangle \otimes \psi$.
- Different orientation conventions differ by the *external* symmetry $U_S(g)$ acting on the reduced description.

1.3 Relational observables

Definition 1.3.1 (Relativisation map):

$$\$_R : \mathcal{B}(\mathcal{H}_S) \to \mathcal{B}(\mathcal{H}_R \otimes \mathcal{H}_S), \qquad \$_R(A) = \sum_g |g\rangle\langle g| \otimes \mathcal{U}_g^S(A).$$

$\$_R(A)$ is the *relational observable* "A, given that R is in orientation e".

Theorem 1.3.1: Relational observables and the reduction map

(i) $\$_R(A)$ is invariant under $U(g)$, hence it preserves $\mathcal{H}_{\text{phys}}$.

(ii) $\$_R$ is a unital $*$-homomorphism.

(iii) $\mathcal{R}_e \, \$_R(A)\big|_{\mathcal{H}_{\text{phys}}} \mathcal{R}_e^{-1} = A$ for all $A \in \mathcal{B}(\mathcal{H}_S)$.

Hence the relational observables restricted to $\mathcal{H}_{\text{phys}}$ form an algebra isomorphic to $\mathcal{B}(\mathcal{H}_S)$: the perspective of an ideal QRF is *complete*.

Proof. (i) $U(h)\$_R(A)U(h)^\dagger = \sum_g |hg\rangle\langle hg| \otimes \mathcal{U}_{hg}(A) = \$_R(A)$.
(ii) $\langle g|g'\rangle = \delta_{gg'}$, and each $\mathcal{U}_g$ is a $*$-automorphism.
(iii) For $\Psi = \mathcal{R}_e^{-1}\psi$,

$$\$_R(A)\Psi = \frac{1}{\sqrt{|G|}} \sum_g |g\rangle \otimes U_S(g) A U_S(g)^\dagger U_S(g)\psi = \mathcal{R}_e^{-1}(A\psi). \qquad \square$$

Proposition 1.3.2 (Coherent vs. incoherent group averaging): For $\psi \in \mathcal{H}_S$, compare

- the physical state $\Psi = \mathcal{R}_e^{-1}\psi \in \mathcal{H}_{\text{phys}}$, obtained by coherent averaging; and
- the invariant density operator $\rho^{\text{inv}} = \mathcal{G}(|e\rangle\langle e| \otimes |\psi\rangle\langle\psi|) = \frac{1}{|G|}\$_R(|\psi\rangle\langle\psi|)$, obtained by incoherent averaging.

Then $\rho^{\text{inv}} \in \mathcal{S}_{\text{inv}} \setminus \mathcal{S}_{\text{phys}}$ in general. Nevertheless, for all $A \in \mathcal{B}(\mathcal{H}_S)$,

$$\langle \Psi, \$_R(A)\Psi\rangle = \text{Tr}\left[\rho^{\text{inv}} \$_R(A)\right] = \langle \psi, A\psi\rangle.$$

So both descriptions assign the same statistics to all relational observables. They differ only on invariant operators outside $\$_R(\mathcal{B}(\mathcal{H}_S))$.

Proof. The first equality follows from Theorem 1.3.1(iii). For the second, use that $\$_R(A)$ is invariant and $\mathcal{G}$ is self-adjoint:

$$\text{Tr}[\rho^{\text{inv}}\$_R(A)] = \text{Tr}\left[(|e\rangle\langle e| \otimes \psi\psi^\dagger)\$_R(A)\right] = \langle \psi, A\psi\rangle. \qquad \square$$

Remark 1.3.3 (Continuous groups):

- *Compact groups.* With $\mathcal{H}_R = L^2(G)$ and normalised Haar measure, invariant vectors $\Psi(g) = U_S(g)\psi$ satisfy $\|\Psi\|^2 = \int dg \, \|\psi\|^2 = \|\psi\|^2$. They are normalisable in $\mathcal{H}_{\text{kin}}$, even though $L^2(G)$ is infinite-dimensional. Only the orientation kets $|g\rangle$ are improper (Dirac-δ), so $\mathcal{R}_e$ is defined through the continuous representative $\Psi(g)$.
- *Non-compact groups* (e.g. translations $\mathbb{R}$). Here $\int dg = \infty$, invariant vectors are not normalisable, and $\mathcal{H}_{\text{phys}}$ needs an induced (rigging / group-averaging) inner product (de la Hamette et al. 2021).
- For time translations, $\mathcal{R}_e$ is Page–Wootters conditioning on a clock reading (Höhn–Smith–Lock).

Chapter 2 Changes of Quantum Reference Frame

2.1 The frame-change operator

Setting. Two ideal QRFs R_1 and R_2 and a system S, with $\mathcal{H}_{\text{kin}} = \mathcal{H}_1 \otimes \mathcal{H}_2 \otimes \mathcal{H}_S$ and the diagonal action $U_1 \otimes U_2 \otimes U_S$. Applying Theorem 1.2.1 with "system" R_2S (respectively R_1S) gives unitaries

$$\mathcal{R}_e^{(1)} : \mathcal{H}_{\text{phys}} \to \mathcal{H}_2 \otimes \mathcal{H}_S, \qquad \mathcal{R}_e^{(2)} : \mathcal{H}_{\text{phys}} \to \mathcal{H}_1 \otimes \mathcal{H}_S.$$

Definition 2.1.1 (QRF transformation): $V_{1\to 2} := \mathcal{R}_e^{(2)} \big(\mathcal{R}_e^{(1)}\big)^{-1} : \ \mathcal{H}_2 \otimes \mathcal{H}_S \to \mathcal{H}_1 \otimes \mathcal{H}_S$.

Theorem 2.1.1: Explicit form of the frame change

$$V_{1\to 2} = \sum_{h\in G} |h^{-1}\rangle_1\langle h|_2 \otimes U_S(h)^\dagger.$$

That is: "invert the relative orientation and apply the inverse of the controlled symmetry to S". $V_{1\to 2}$ is unitary and $V_{2\to 1} = V_{1\to 2}^\dagger$.

Proof. Let $\phi = \sum_h |h\rangle_2 \otimes \chi_h$. By Theorem 1.2.1,

$$\big(\mathcal{R}_e^{(1)}\big)^{-1}\phi = \frac{1}{\sqrt{|G|}} \sum_{g,h} |g\rangle_1 |gh\rangle_2 \, U_S(g)\chi_h.$$

Projecting onto $\langle e|_2$ selects $gh = e$, i.e. $g = h^{-1}$. Multiplying by $\sqrt{|G|}$,

$$V_{1\to 2}\phi = \sum_h |h^{-1}\rangle_1 \otimes U_S(h^{-1})\chi_h.$$

Unitarity holds because V is a composition of unitaries. Finally, $V_{2\to 1} = \mathcal{R}_e^{(1)}(\mathcal{R}_e^{(2)})^{-1} = V_{1\to 2}^{-1}$. □

Proposition 2.1.2 (Composition): For three ideal frames, $V_{1\to 3} = V_{2\to 3} \circ V_{1\to 2}$ as maps between the corresponding reduced spaces.

Proof. $\mathcal{R}_e^{(3)}(\mathcal{R}_e^{(2)})^{-1}\mathcal{R}_e^{(2)}(\mathcal{R}_e^{(1)})^{-1} = \mathcal{R}_e^{(3)}(\mathcal{R}_e^{(1)})^{-1}$. □

Proposition 2.1.3 (Transformation of observables): If X is an operator on $\mathcal{H}_2 \otimes \mathcal{H}_S$ in the perspective of R_1, the same physical observable in the perspective of R_2 is $V_{1\to 2} X V_{1\to 2}^\dagger$. Expectation values agree: $\langle \phi, X\phi\rangle = \langle V\phi, VXV^\dagger V\phi\rangle$.

2.2 Frame dependence of superposition and entanglement

Corollary 2.2.1: Let $\phi = \big(\sum_h c_h |h\rangle_2\big) \otimes |\chi\rangle_S$ be a product state in the perspective of R_1. In the perspective of R_2,

$$V_{1\to 2}\phi = \sum_h c_h \, |h^{-1}\rangle_1 \otimes U_S(h)^\dagger|\chi\rangle.$$

- If R_2 is sharply oriented ($c_h = \delta_{h,h_0}$), the result is again a product state.
- Otherwise it is *entangled* between R_1 and S, unless the vectors $U_S(h)^\dagger|\chi\rangle$ for $h \in \text{supp}(c)$ are all collinear.

Proof. Theorem 2.1.1. The vectors $|h^{-1}\rangle$ are orthonormal, so the Schmidt rank equals the dimension of the span of $\{U_S(h)^\dagger\chi\}_{h\in \text{supp}\, c}$. □

Remark 2.2.2:
- Superposition and entanglement are *perspective-dependent* notions (Giacomini–Castro-Ruiz–Brukner): a frame in superposition relative to R_1 "sees" R_1 in superposition, correlated with S.
- Relatedly, how the total system splits into subsystems is itself frame-relative (de la Hamette et al. 2021).
- The operational framework of Chapter 3 shows that part of this entanglement is not accessible to relational observables, which weakens claims of physical entanglement generation by frame changes (Carette–Głowacki–Loveridge).

Remark 2.2.3 (Continuous groups): For $G = (\mathbb{R}, +)$ acting by translations, Theorem 2.1.1 becomes the operator of Giacomini–Castro-Ruiz–Brukner,

$$V = \mathcal{P}_{12}\, e^{\frac{i}{\hbar}\hat{x}_2\hat{p}_S},$$

a parity-swap times a controlled translation (formal, with non-normalisable orientation states). Boosts and general locally compact groups are treated by de la Hamette–Galley.

Chapter 3 Operational Quantum Reference Frames

3.1 Frames as covariant POVMs

Definition 3.1.1 (Operational QRF): A QRF is a triple $\mathsf{R} = (\mathcal{H}_R, U_R, \mathsf{E}_R)$, where $\mathsf{E}_R = \{\mathsf{E}_R(g)\}_{g\in G}$ is a POVM that is *covariant*:

$$U_R(h)\,\mathsf{E}_R(g)\,U_R(h)^\dagger = \mathsf{E}_R(hg).$$

E_R is the *frame observable* ("which orientation is the frame in?").

- The frame is *sharp* if E_R is projection-valued.
- The ideal QRF of Chapter 1 is sharp, with $\mathsf{E}_R(g) = |g\rangle\langle g|$.

Definition 3.1.2 (Relativisation): $¥_{\mathsf{R}}(A) = \sum_g \mathsf{E}_R(g) \otimes \mathcal{U}_g^S(A)$ for $A \in \mathcal{B}(\mathcal{H}_S)$.

Theorem 3.1.1: Properties of relativisation

(i) $¥_{\mathsf{R}}$ is unital and completely positive, and its image consists of invariant operators.

(ii) If R is sharp, then $¥_{\mathsf{R}}$ is a $*$-homomorphism.

(iii) For product states, $\mathrm{Tr}[(\omega\otimes\rho)\,¥_{\mathsf{R}}(A)] = \mathrm{Tr}[\Gamma_\omega(\rho)\,A]$, where

$$\Gamma_\omega(\rho) = \sum_g \mu_\omega(g)\,U_S(g)^\dagger \rho\, U_S(g), \qquad \mu_\omega(g) = \mathrm{Tr}[\omega\,\mathsf{E}_R(g)].$$

Proof. (i) Unital: $\sum_g \mathsf{E}(g) = \mathbb{I}$. CP: each term $A \mapsto \mathsf{E}(g)\otimes U(g)AU(g)^\dagger$ is CP because $\mathsf{E}(g) \geqslant 0$. Invariance:

$$(U_R(h)\otimes U_S(h))\,¥(A)\,(\cdot)^\dagger = \sum_g \mathsf{E}(hg)\otimes\mathcal{U}_{hg}(A) = ¥(A).$$

(ii) $\mathsf{E}(g)\mathsf{E}(g') = \delta_{gg'}\mathsf{E}(g)$.
(iii) $\mathrm{Tr}[(\omega\otimes\rho)¥(A)] = \sum_g \mu_\omega(g)\,\mathrm{Tr}[\rho\,U(g)AU(g)^\dagger]$; now use cyclicity of the trace. □

Corollary 3.1.3 (Absolute description as a localisation limit):
- If the frame state is sharply localised at the identity ($\mu_\omega = \delta_e$), then relative statistics equal absolute statistics: $\Gamma_\omega = \mathrm{id}$.

- In general, S relative to a quantum frame is described by a weighted twirl with the frame's orientation distribution μ_ω. A *delocalised quantum frame* acts like a *partially shared external frame* (Section 0.2).

Remark 3.1.4:
- For compact Lie groups and finite-dimensional R, no covariant PVM on G exists; frames are necessarily unsharp. Relative observables then approximate absolute ones only as the frame becomes large and localised: an *accuracy–size trade-off* (Miyadera–Loveridge–Busch).
- For non-ideal frames, reduction maps and frame changes are generally *not unitary*. Operational frame changes act on equivalence classes of relative states (Carette–Głowacki–Loveridge). Truncated or irreversible frame changes are treated by de la Hamette–Galley.

Chapter 4 Quantum Reference Frames as Resources

QRF framework vs. resource theory of asymmetry. The two are related, but they are not the same theory.

	QRF framework (Ch. 1–3)	**Asymmetry QRT (this chapter)**
Question	How do states and observables depend on the chosen frame?	Without a frame, which states can stand in for one, and how well?
Objects	frames, relational states, relational observables	invariant states $\mathcal{S}_{\text{inv}}$, covariant channels
Structure	reduction maps, frame changes, relativisation	free operations, convertibility, monotones
Typical result	frame-dependence of entanglement	$R_{\text{rel}} = S(\mathcal{G}\rho) - S(\rho)$

The link: a state that is a resource in the asymmetry QRT is precisely a state that can serve as a (bounded) quantum reference frame.

4.1 Lifting superselection rules

Theorem 4.1.1: SSRs can be lifted by an ideal QRF

For $\rho^{\text{inv}} = \$_R(\rho)/|G| \in \mathcal{S}_{\text{inv}}$ (incoherent averaging; not in $\mathcal{S}_{\text{phys}}$ in general, cf. Proposition 1.3.2), every effect E and every Kraus family $\{K_i\}$ on S:

- $\$_R(E)$ and $\{\$_R(K_i)\}$ are an invariant effect and an invariant Kraus family on RS;
- $\mathrm{Tr}[\rho^{\text{inv}}\,\$_R(E)] = \mathrm{Tr}[\rho E]$.

Proof. Invariance and the homomorphism property come from Theorem 1.3.1, which gives $\sum_i \$(K_i)^\dagger \$(K_i) = \$(\mathbb{I}) = \mathbb{I}$. For the statistics,

$$\mathrm{Tr}[\$(\rho)\$(E)] = \mathrm{Tr}[\$(\rho E)] = \sum_g \mathrm{Tr}[\mathcal{U}_g(\rho E)] = |G|\ \mathrm{Tr}[\rho E]. \qquad \square$$

Remark 4.1.1:
- SSRs from compact groups are therefore not fundamental.
- Cheaters can replace SSR-violating operations V by $\$(V)$ (Kitaev–Mayers–Preskill), so security based on SSRs is conditional on the adversary lacking a frame.

4.2 The resource theory of asymmetry

Theorem 4.2.1: Asymmetry as a QRT

With free states = invariant states and free operations = covariant channels:

(i) this is a QRT with tensor-product structure;

(ii) $\mathcal{O}$ is affine;

(iii) $\mathcal{G}$ is a linear CPTP resource-destroying map.

Its resources are exactly the states that can serve as (bounded) quantum reference frames.

Proof. Covariance is preserved under composition, $\otimes$id, appending invariant states and partial trace. It is a linear condition, so (ii) follows. For (iii), $\mathcal{G}(\rho)$ is invariant and $\mathcal{G}$ fixes invariant states. □

Theorem 4.2.2: Relative entropy of frameness

$$\min_{\sigma \text{ inv.}} S(\rho\|\sigma) = S(\rho\|\mathcal{G}(\rho)) = S(\mathcal{G}(\rho)) - S(\rho).$$

Proof. For invariant σ, $\log\sigma$ is invariant, so $\mathrm{Tr}[\rho\log\sigma] = \mathrm{Tr}[\mathcal{G}(\rho)\log\sigma]$ (self-adjointness of $\mathcal{G}$). Hence

$$S(\rho\|\sigma) = \big[S(\mathcal{G}\rho) - S(\rho)\big] + S(\mathcal{G}\rho\|\sigma),$$

which is minimal at $\sigma = \mathcal{G}\rho$. □

Remark 4.2.1:

- *Non-extensivity.* Let $U(\theta) = e^{i\theta\hat{Q}}$ with integer charges in $[q_{\min}, q_{\max}]$, and put $\Delta q := q_{\max} - q_{\min}$. On n copies the total charge takes at most $K = n\Delta q + 1$ consecutive integer values. Then $\mathcal{G}$ is the pinching onto these sectors, which equals the uniform mixture $\frac{1}{K}\sum_{k=0}^{K-1} Z^k(\cdot)Z^{-k}$ with $Z = e^{2\pi i\hat{Q}_{\mathrm{tot}}/K}$. A uniform mixture of K unitary conjugations raises entropy by at most $\log K$, so

$$S(\mathcal{G}(\rho^{\otimes n})) - S(\rho^{\otimes n}) \leqslant \log(n\Delta q + 1), \qquad \text{hence} \qquad \lim_n \tfrac{1}{n}\big[S(\mathcal{G}(\rho^{\otimes n})) - S(\rho^{\otimes n})\big] = 0.$$

  (With Δq the bound is sharp at $n = 1$: for charges $-1, 0, 1$ and $\psi \propto |-1\rangle + |0\rangle + |1\rangle$ the left side is $\log 3$.) For every compact Lie group, the regularised quantity vanishes (Gour–Marvian–Spekkens).

- *Conservation:* asymmetry monotones are conserved by covariant unitaries, which extends Noether's theorem (Marvian–Spekkens).

4.3 Communicating and consuming frames

Theorem 4.3.1: Optimal frame alignment (maximum likelihood)

Alice sends $U(g)|\psi\rangle$, with g uniformly distributed, and Bob uses the covariant POVM

$$E_g = \frac{1}{|G|}\,U(g)|e\rangle\langle e|U(g)^\dagger, \qquad \sum_g E_g = \mathbb{I}_{\widetilde{\mathcal{H}}},$$

on the orbit space

$$\widetilde{\mathcal{H}} = \bigoplus_q \mathcal{M}_q \otimes \widetilde{\mathcal{N}}_q, \qquad \dim\widetilde{\mathcal{N}}_q = r_q := \min(d_q, \dim\mathcal{N}_q).$$

Then:

- the *likelihood density* (with respect to the normalised Haar measure) of the correct guess is $\mu = |\langle e|\psi\rangle|^2$, with maximum $\mu_{\max} = \dim \widetilde{\mathcal{H}} = \sum_q d_q r_q$;
- the *success probability* is $P_{\text{succ}} = \mu/|G|$, so

$$P_{\text{succ}}^{\max} = \frac{\dim \widetilde{\mathcal{H}}}{|G|} \leqslant 1.$$

Proof. Completeness. $\sum_g E_g = \mathcal{G}(|e\rangle\langle e|)$, so completeness means $\mathcal{G}(|e\rangle\langle e|) = \mathbb{I}_{\widetilde{\mathcal{H}}}$. By Theorem 0.1.1, each $\Pi_q|e\rangle$ must then be $\sqrt{d_q}$ times a maximally entangled vector of rank r_q, so $\|e\|^2 = \sum_q d_q r_q$.

Success probability. For each g, $\mathrm{Tr}[E_g U(g)|\psi\rangle\langle\psi|U(g)^\dagger] = |\langle e|\psi\rangle|^2/|G|$; averaging over the uniform prior gives $P_{\text{succ}} = \mu/|G|$.

Bound. Cauchy–Schwarz gives $\mu \leqslant \|e\|^2$, with equality iff $\psi \propto e$. Finally, $\sum_q d_q r_q \leqslant \sum_q d_q^2 \leqslant |G|$, since each irrep q occurs d_q times in the regular representation; so $P_{\text{succ}}^{\max} \leqslant 1$, as it must be. □

Remark 4.3.1:

- *Scaling (model dependent).* For specific models, the optimal error reaches the Heisenberg scaling $\sim 1/N^2$ with N systems, versus $1/N$ for unentangled signals. Examples are a $U(1)$ phase reference with at most N photons and an $SU(2)$ direction or Cartesian frame with N spins, each with fidelity-type figures of merit, entangled signal states and collective covariant measurements (Bartlett–Rudolph–Spekkens). This is *not* a consequence of Theorem 4.3.1, which concerns maximum likelihood.

- *Reusability depends on the group.* For finite G, an ideal frame in state $|e\rangle$ is perfectly reusable: $\$_R(V)(|e\rangle \otimes \psi) = |e\rangle \otimes V\psi$ for every unitary V on S. For continuous symmetries, finite-dimensional frames cannot perfectly distinguish orientations. They can then suffer unavoidable degradation in some tasks; e.g. a spin-j reference direction allows $\sim j^2$ uses in a specific measurement model (Bartlett–Rudolph–Spekkens).

- Shared asymmetric states ("refbits") activate entanglement that is hidden under a local SSR (Bartlett–Rudolph–Spekkens).

Chapter 5 Outlook

- *Time as a QRF:* the Page–Wootters formalism, relational Schrödinger dynamics and clock changes are the special case $G = \mathbb{R}$ (time translations) of Chapters 1–2; the "trinity" of Höhn–Smith–Lock shows the equivalence of the main approaches.

- *Gauge theory and gravity:* the perspective-neutral space is the constraint surface of a gauge system, so QRFs provide gauge-invariant relational observables. This connects to crossed products in algebraic QFT and gravity.

- *Open problems:* frame changes for non-ideal frames; frames for non-compact groups and the role of mass; operational meaning of frame-dependent entanglement.

Further Reading

A. Foundations and the external-frame picture (Chapter 0, 4)

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B. Frame changes and the perspective-neutral picture (Chapters 1–2)

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C. Operational quantum reference frames (Chapter 3)

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- P. A. Höhn, A. R. H. Smith, M. P. E. Lock, *Trinity of relational quantum dynamics*, Phys. Rev. D **104**, 066001 (2021). arXiv:1912.00033

E. Frames as resources (Chapter 4)

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